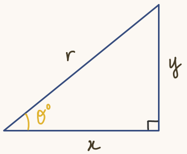
PURE 5.

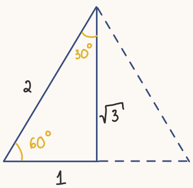
Trigonometry

1 Angles between 0° and 90°

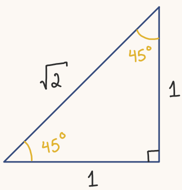


$\sin\theta = \dfrac{\text{opposite}}{\text{hypotenuse}}$ $\cos\theta = \dfrac{\text{adjacent}}{\text{hypotenuse}}$ $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$

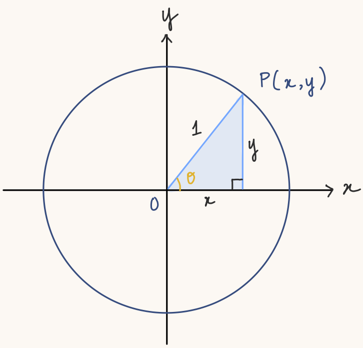
$\sin\theta = \dfrac{y}{r}$ $\cos\theta = \dfrac{x}{r}$ $\tan\theta = \dfrac{y}{x}$



$\sin 30^\circ = \frac{1}{2}$	$\sin 60^\circ = \frac{\sqrt{3}}{2}$	$\sin 45^\circ = \frac{1}{\sqrt{2}}$
$\cos 30^\circ = \frac{\sqrt{3}}{2}$	$\cos 60^\circ = \frac{1}{2}$	$\cos 45^\circ = \frac{1}{\sqrt{2}}$
$\tan 30^\circ = \frac{1}{\sqrt{3}}$	$\tan 60^\circ = \sqrt{3}$	$\tan 45^\circ = 1$



* the unit circle



- center: O (origin)
- radius = 1 unit
- for any acute angle between OP and x-axis:

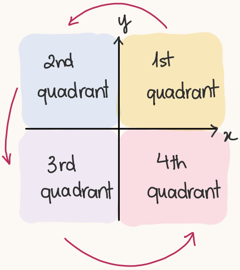
a.k.a. basic angle / reference angle

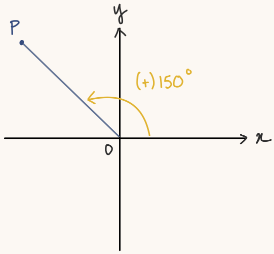
$\sin\theta = \dfrac{y}{1} = y$ $\sin 0^\circ = 0$

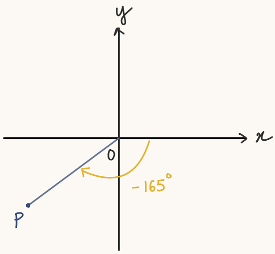
$\cos\theta = \dfrac{x}{1} = x$ $\cos 0^\circ = 1$

$\tan\theta = \dfrac{y}{x}$

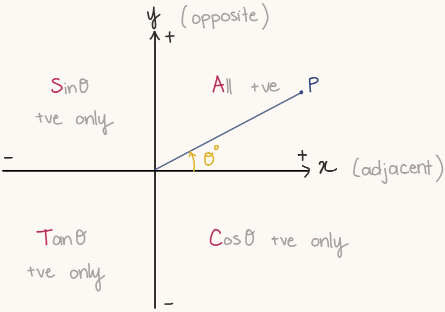
2 Angles : calculated anti-clockwise from the x-axis = +ve value







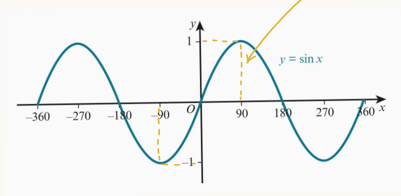
- use a CAST diagram to decide whether sin, cos or tan θ is +ve or -ve



3. Trigonometric graphs

- $y = \sin\theta$

amplitude : 1

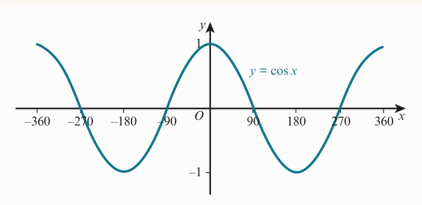


- period : 360° (repeats every 360°)
- rotational order of 2 about the origin (in a 360°, it looks the same 2 times)
- $-1 \leq \sin\theta \leq 1$

minimum maximum

periodic functions

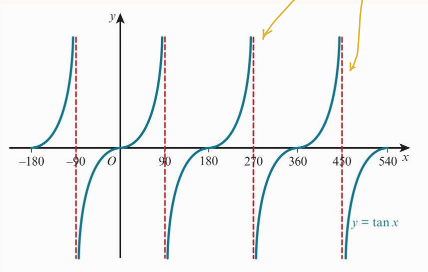
- $y = \cos x$



- period : 360°
- symmetrical about y-axis
- $-1 \leq \sin\theta \leq 1$

asymptotes : graph gets infinitely close but never touches these lines

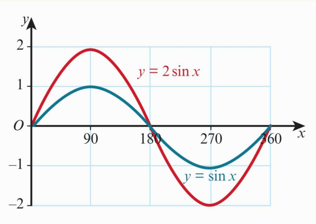
· $y = \tan\theta$



· period : 180°
· rotational symmetry 2 about origin
· $\tan\theta$ can be any real number
  · as $\theta \rightarrow 90°$, $\tan\theta \rightarrow \infty$
  · as $\theta \rightarrow -90°$, $\tan\theta \rightarrow -\infty$

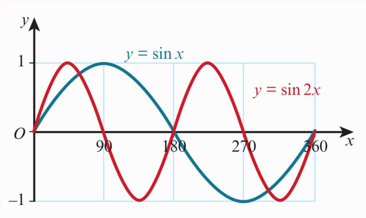
× trig graph transformations

· $y = a\sin x$



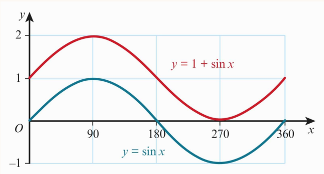
- stretch ; factor 2
- // y - axis
- period : 360°

· $y = \sin ax$



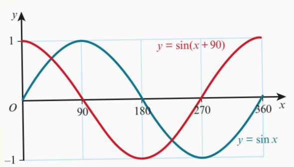
- stretch ; factor $\frac{1}{2}$ (= squashed)
- // x - axis
- period · 180°

· $y = a + \sin x$



- translation ; of $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- amplitude · 1
- period : 360°

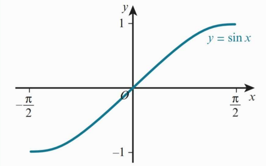
· $y = \sin(x + a)$



- translation ; of $\begin{pmatrix} -90 \\ 0 \end{pmatrix}$
- amplitude · 1
- period : 360°

4. Inverse trig functions

- $y = \sin x$

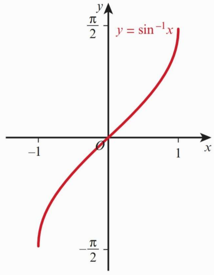


$y = \sin x$

domain: $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$

range: $-1 \leqslant \sin x \leqslant 1$

- $y = \sin^{-1} x$

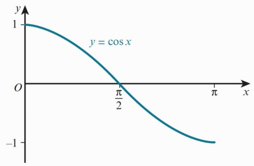


$y = \sin^{-1} x$

domain: $-1 \leqslant x \leqslant 1$

range: $-\frac{\pi}{2} \leqslant \sin^{-1} x \leqslant \frac{\pi}{2}$

- $y = \cos x$

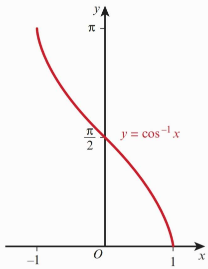


$y = \cos x$

domain: $0 \leqslant x \leqslant \pi$

range: $-1 \leqslant \cos x \leqslant 1$

- $y = \cos^{-1} x$

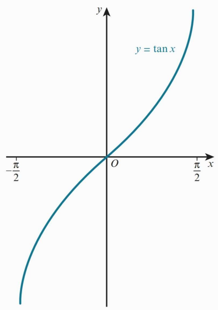


$y = \cos^{-1} x$

domain: $-1 \leqslant x \leqslant 1$

range: $0 \leqslant \cos^{-1} x \leqslant \pi$

- $y = \tan x$

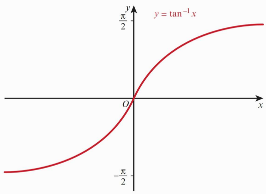


$y = \tan x$

domain: $-\frac{\pi}{2} < x < \frac{\pi}{2}$

range: $\tan x \in \mathbb{R}$

- $y = \tan^{-1} x$



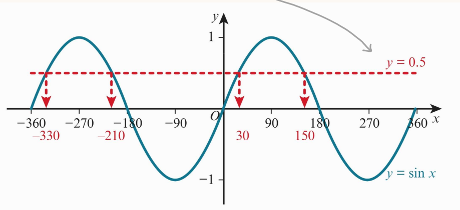
$y = \tan^{-1} x$

domain: $x \in \mathbb{R}$

range: $-\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2}$

5. Trigonometric equations

Ex: solve $\sin x = 0.5$ for $-360° \leq x \leq 360°$

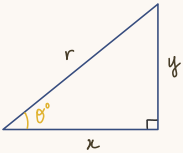


- use calculator to find $x = 30°$

↳ use symmetry of graph to find remaining values

⟶ 4 values total

*** Rule 1.**



$$\left.\begin{aligned} \sin\theta &= \frac{y}{r} \\ \cos\theta &= \frac{x}{r} \end{aligned}\right\} \quad \tan\theta = \frac{y}{x} = \frac{\left(\frac{y}{r}\right)}{\left(\frac{x}{r}\right)} \longrightarrow \boxed{\tan\theta = \frac{\sin\theta}{\cos\theta}}$$

for all θ, with $\cos\theta \neq 0$

$\equiv$ is an identity

= true for ALL values of x

$$\boxed{\tan x \equiv \frac{\sin x}{\cos x}}$$

*** Rule 2**

$$x^2 + y^2 = r^2$$

$$\left(\frac{x}{r}\right)^2 + \left(\frac{y}{r}\right)^2 = 1$$

$$\longrightarrow \boxed{\cos^2\theta + \sin^2\theta = 1}$$ for all θ.

$$\boxed{\sin^2 x + \cos^2 x \equiv 1}$$